THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Bhaskara Contest

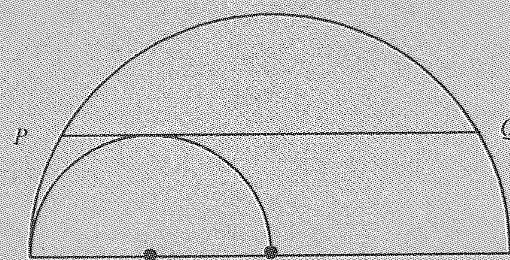
(NMTC- at ***JUNIOR LEVEL*** *IX & X Standards)*

Saturday, 30^{th} August 2008.

Note:

1) *Fill in the answer sheet with your Name, Class, and Section etc. in the specified places.*
2) *Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).*
3) *For each correct response you get 1 mark;* ***for each incorrect response you lose ¼ mark.***
4) *Diagrams are only visual aids; they are not drawn to scale.*
5) *You are free to do rough work on separate sheets.*
6) *Duration of the test: 2 p.m. to 4. p.m.- 2 hours*

Two semi circles are constructed as shown in the figure. The chord PQ of the greater circle touches the smaller circle as in parallel to the diameter of larger circle. If the length of PQ is 10 units, then what is the area between the semi circles?



(A) 50 **(B)** $50\,\pi$ **(C)** $\frac{1}{2}.25\,\pi$ **(D)** $\frac{1}{2}.25$

	1	2	3	4	5	6	1	2
	6	3	4	5	6	1	2	3
	5	2	1	2	3	4	3	↓
	4	1	6	1	2	5	4	
	3	6	5	4	3	6	5	
	2	5	4	3	2	1	6	
	1	6	5	4	3	2	1	

The digits 1,2,3,4,5 and 6 are written in a spiral like fashion beginning from the central marked cell. Which digit is written on the cell placed exactly 100 cells above the marked cell?

(A) 1 **(B)** 3 **(C)** 5 **(D)** 6

An island is inhabited by both liars and knights. Every knight always tells the truth and each liar always lies. Once day 12 islanders gathered togather and issued a few statements. Two people said "Exactly two of us are liars". Another four people said "Exactly four of us are liars". The remaining six said "Exactly six of us are liars". How many liars are there?

(A) 10 **(B)** 8 **(C)** 6 **(D)** 4

4. For how many real values of α will $x^2 + 2\alpha x + 2008 = 0$ have two integer roots?

5. The first digit from the left of a 4 digit number is equal to the number of zeroes in the number. The second digit is equal to the number of digits 1, the third digit is equal to the number of digits 2 and the fourth digit is equal to the number of digits 3. How many numbers have this property?

(A) 0 **(B)** 2 **(C)** 3 **(D)** 4

6. Let A be the least number such that $10A$ is a perfect square and $35A$ is a perfect cube. Then the number of positive divisors of A is

(A) 72 **(B)** 64 **(C)** 45 **(D)** 80

7. The number of positive integer solutions of $ab - 24 = 2a$ is ?

(A) 0 **(B)** 2 **(C)** 8 **(D)** 12

8. What is the least possible value of the expression $2008 - BHA - SK - ARA$ if it is known that each alphabet represents a different non zero digit?

(A) 106 **(B)** 108 **(C)** 1580 **(D)** none of these

9. In one move the king can go to any adjacent square, along a row, column or diagonal. How many routes with the minimum number of moves are there for the king to travel from the top left square to the bottom right square?

K				
				K

(A) 35 **(B)** 2 **(C)** 15 **(D)** 4

10. The sum of $\frac{1}{2\sqrt{1}+1\sqrt{2}}+\frac{1}{3\sqrt{2}+2\sqrt{3}}+\frac{1}{4\sqrt{3}+3\sqrt{4}}+\ldots\ldots\ldots\ldots+\frac{1}{25\sqrt{24}+24\sqrt{25}}$ is

(A) $\frac{9}{10}$ **(B)** $\frac{4}{5}$ **(C)** $\frac{14}{15}$ **(D)** $\frac{7}{15}$

11. The number of pairs of prime numbers (p, q) satisfying the condition $\frac{51}{100}<\frac{1}{p}+\frac{1}{q}<\frac{5}{6}$ will be

(A) 49 **(B)** 24 **(C)** 50 **(D)** 48

12. A positive integer 'n' is such that its only digits are 3's . 'n' is exactly divisible by 383 when $\frac{n}{383}$ is divided by 1000, what is the remainder?

(A) 351 **(B)** 781 **(C)** 651 **(D)** 931

3. Here is a Morse – Thue sequence. Start with 0; to each initial segment append its complement, as follows; 0, 01, 0110, 01101001 . . . What will be the last 8 digits of the 2008th term of this sequence.

(A) 11110000 **(B)** 10010110 **(C)** 01101001 **(D)** 00001111

4. The number of 2 digit numbers that has exactly 6 factors is

(A) 4 **(B)** 20 **(C)** 16 **(D)** 12

5. $P = 2008^{2007} - 2008$; $Q = 2008^2 + 2009$. The remainder when P is divided by Q is

(A) 4031964 **(B)** 4032066 **(C)** 4158972 **(D)** 40682896

6. ABC is a right angled triangle with $\angle A = 63^0$ $\angle B = 27^0$ $ACFG$, $BCDE$ are squares drawn external to the triangle ABC. Such that AE intersects BC at H, BG intersects AC at K. Then the size of angle measure DKH is

(A) 27^0 **(B)** 36^0 **(C)** 63^0 **(D)** 45^0

7. If $f\ (x) + f\ (1-x)$ is equal to 10 for all real numbers x then $f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + f\left(\frac{3}{100}\right) + \dots\dots\dots\dots + f\left(\frac{99}{100}\right)$ equals

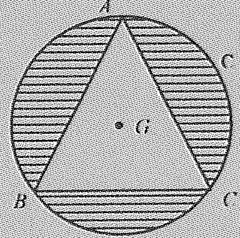
(A) 490 **(B)** 495 **(C)** 500 **(D)** 505

8. A teacher wrote the numbers 1,2,3, . . . 2008 on the black board; one of the students erases m of these numbers. Among the numbers remaining on the board, not a single one is the product of any two of the others. If the smallest possible value of m is x then

(A) $x < 50$ **(B)** $50 \le x < 200$ **(C)** $200 \le x < 500$ **(D)** $500 \le x < 1000$

9. x,y are positive real number satisfying the system $x^2 + y\sqrt{xy} = 336$, $y^2 + x\sqrt{xy} = 112$ then $x + y$ equals

(A) $\sqrt{448}$ **(B)** $\sqrt{224}$ **(C)** 20 **(D)** 40

20. 

ABC is an equilateral Δ. G is the centroid and $AG = 2$cm. C is the circum circle of the triangle. The area of the shaded portion (in cm^2) is

(A) $(4\pi - 3\sqrt{3})$ **(B)** $\dfrac{16\pi - 9\sqrt{3}}{4}$ **(C)** $3\,(4\pi - 3\sqrt{3})$ **(D)** $\frac{1}{3}(4\pi - 3\sqrt{3})$

21. The number of integer solutions of the equation $2^x\,(4 - x) = 2x + 4$ is

22. If a, b, c are positive integers such that $a^2 + 2b^2 - 2ab = 169$ and $2ab - c^2 = 169$ then $a + b + c$ is

(A) 0 **(B)** 169 **(C)** 13 **(D)** 39

23. Two of the altitudes of a scalene triangle have length 4 and 12. If the length of the remaining altitude is also an integer, then its maximum value is

(A) 1 **(B)** 2 **(C)** 5 **(D)** 8

24. Three faces of a rectangular parallelepiped have areas 48cm^2, 18cm^2 and 24cm^2. Then the sum of the length, breadth and height of the parallelepiped (in cm) is

(A) 17 **(B)** 27 **(C)** 14 **(D)** 11

25. $ABCD$ is a square. With centres B and C and radius equal to the side of the square circles are drawn to cut one another at E inside the square. $\angle BDE$ is equal to

(A) $22\frac{1}{2}^0$ **(B)** 30^0 **(C)** 15^0 **(D)** $37\frac{1}{2}^0$

26. p and q are primes. It is given that the quadratic equation $x^2 - px + q = 0$ has distinct real roots. Then $(p + 2q)$ is

(A) 7 **(B)** 19 **(C)** 9 **(D)** 13

27. The number of two-digit numbers that increase by 75% when their digits are reversed is

(A) 3 **(B)** 4 **(C)** 5 **(D)** 7

28. Arun thought of a positive integer. Balu multiplied this by either 5 or by 6. Chitra added 5 or 6 to Balu's number. Devi subtracted either 5 or 6 from the number got by Chitra and the final result was 73. Then the numbers obtained by Arun, Balu and Chitra are respectively.

(A) 12, 72, 78 **(B)** 12,60,66 **(C)** 12,72,77 **(D)** 12,60, 65

29. AB is a fixed diameter of a circle. PQ is a chord whose length is equal to the radius of the circle, which does not intersect the diameter AB. AP and BQ meet at R

(A) $\angle ARB = 45^o$ when PQ is parallel to AB.
(B) $\angle ARB = 60^o$ for all positions of PQ
(C) $\angle ARB = 60^o$ only when PQ is parallel to AB
(D) $\angle ARB$ is a variable angle depending on the position of PQ.

30. a, b are positive integers. Given that $\log_9 a = \log_{12} b = \log_{16}(a + b)$ then the value of $\left(\frac{b}{a}\right)$ is